AI MECH LAB

Mechatronics Course — Beginner Stage

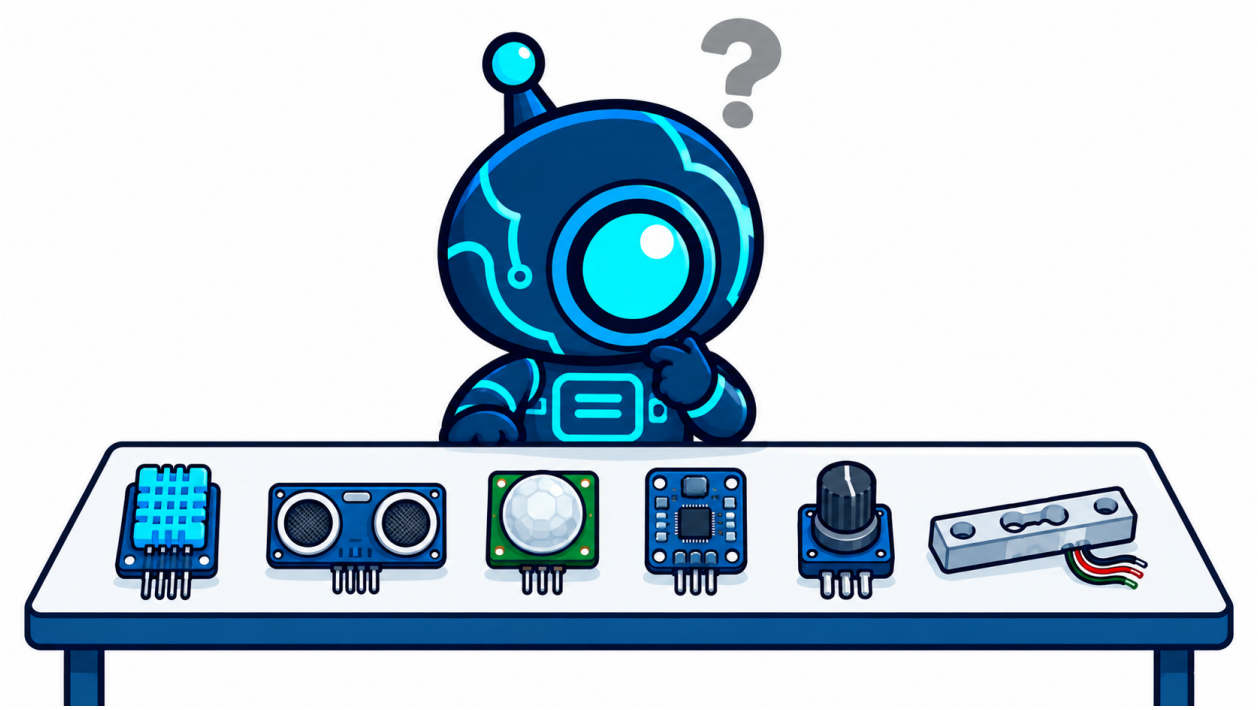


Module 3 · Topic 3.2

Common Sensor Types

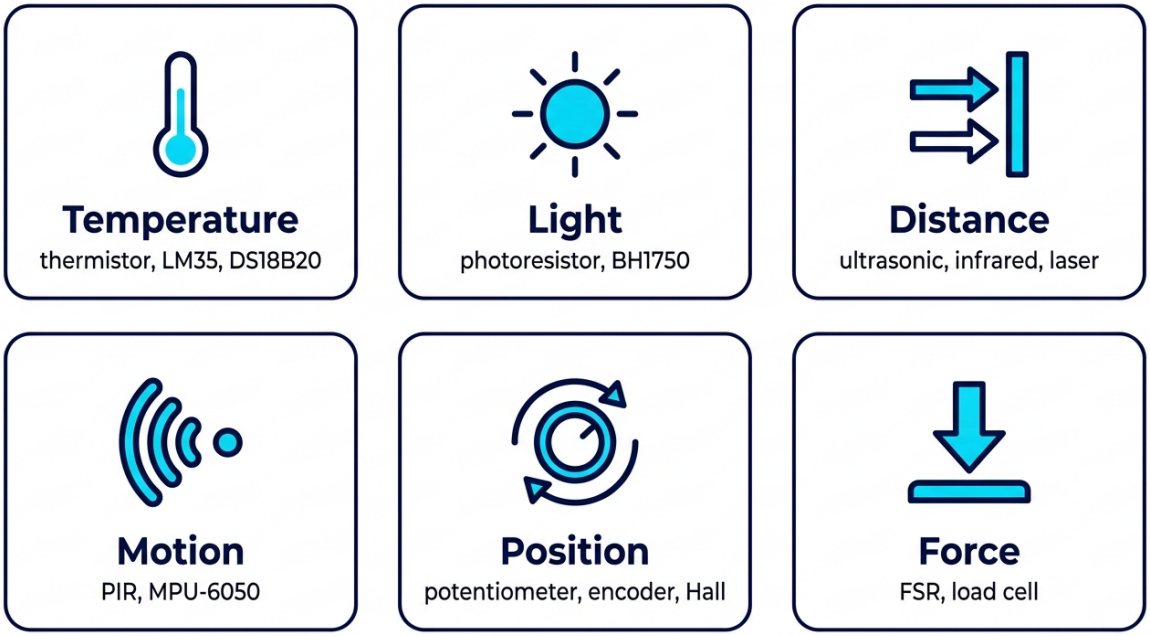
Common Sensor Types

Open any supplier's catalogue and you will find hundreds of sensors. They look different, but they sort quickly once you ask two questions. What physical quantity does it measure? And what kind of signal does it give back? The first question tells you which family it belongs to. The second, from Topic 3.1, tells you how to wire it and read it.



Hundreds of sensors, six families. The trick is knowing which question to ask first.

This topic covers six families you will meet constantly: temperature, light, distance, motion, position and force. Each comes with one or two common parts so the ideas stay concrete. The part names are examples of each type, not recommendations. When you build, the datasheet for the exact part in your hand is the final word.

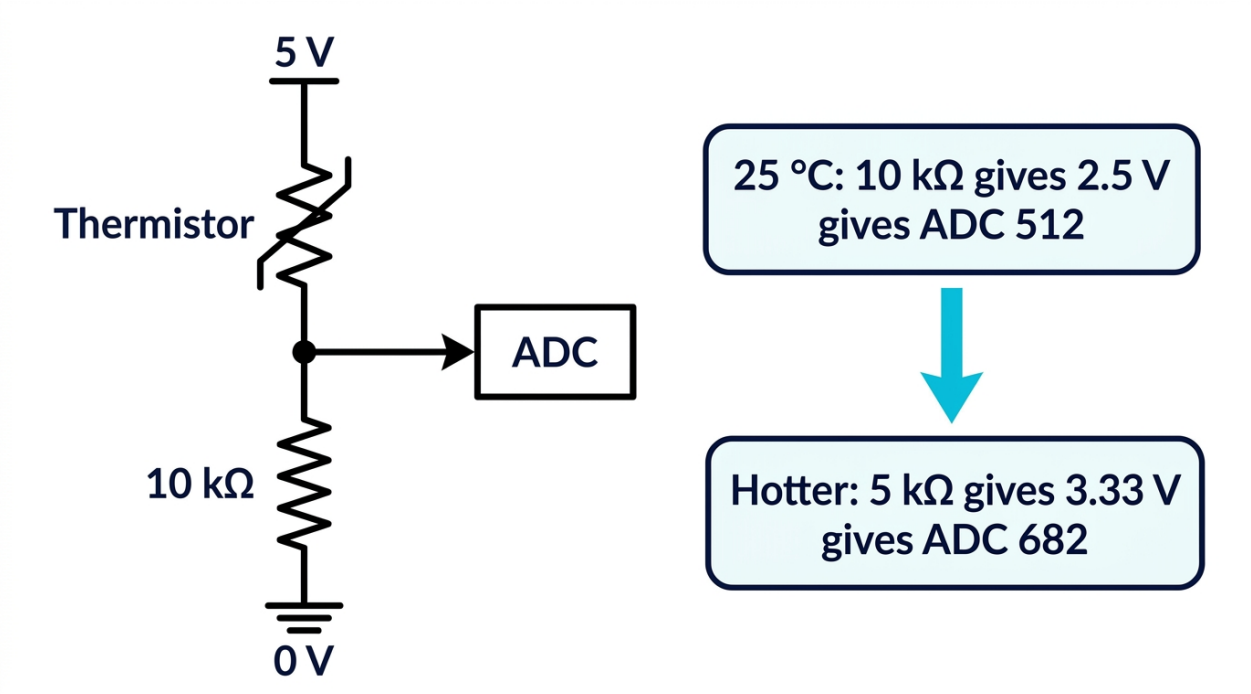


The six sensor families and a few common parts from each.

Resistive sensors: a note before we start

Several sensors never output a voltage at all. A thermistor, a photoresistor and a force-sensitive resistor all change their resistance. A microcontroller reads voltage, not resistance, so you convert one into the other with a voltage divider, the same circuit you met in Module 2.

Put a thermistor in series with a 10 kΩ fixed resistor across 5 V, and read the voltage at the junction between them. A “10k” thermistor is 10 kΩ at 25 °C, so at room temperature the junction sits at 2.5 V and a 10-bit ADC reports 512. Warm it up until its resistance falls to 5 kΩ and the junction rises to 3.33 V, an ADC reading of 682. The resistance changed, the divider turned that into a voltage, and the ADC turned the voltage into a number. Swap the two resistors and the voltage moves the opposite way, so always work out which direction your wiring gives.



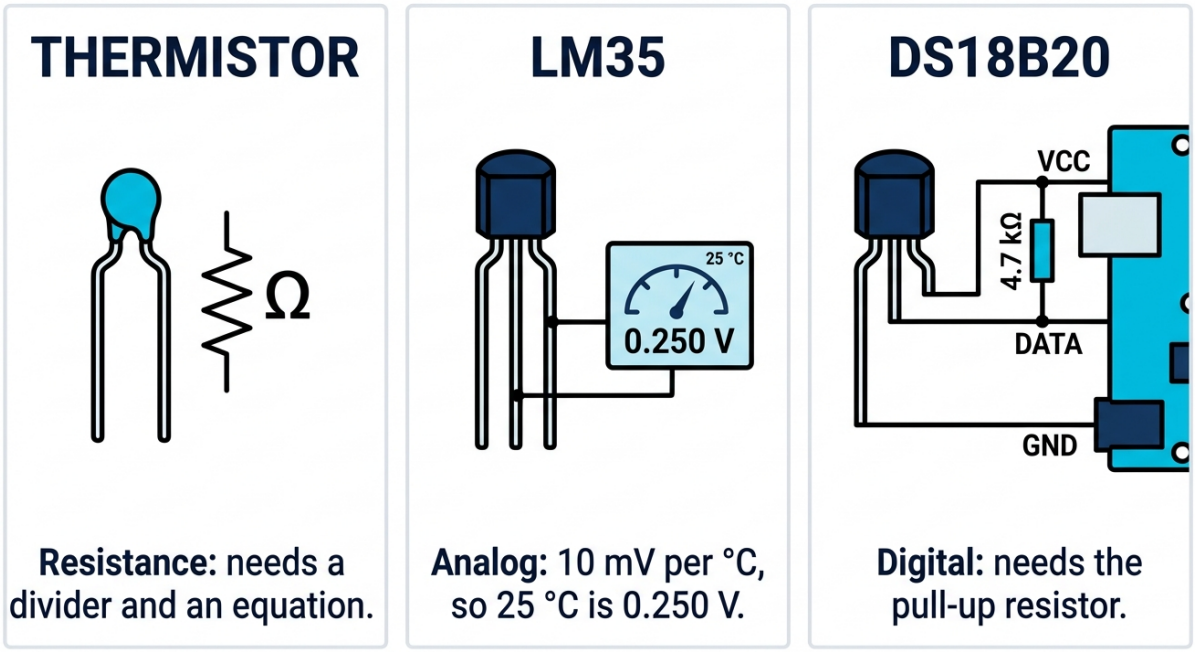
A thermistor in a voltage divider: resistance becomes voltage, and the ADC turns voltage into a number.

Temperature

The **NTC thermistor** is cheap, small and fast. NTC means its resistance falls as temperature rises. The catch is that the change is not a straight line, so your software converts resistance to temperature with an equation from the datasheet, usually the B-parameter equation. Thermistors that are both labeled “10k” can carry different constants depending on the manufacturer, so use the numbers for your exact part.

The **LM35** gives an analog voltage of 10 mV per °C, so 25 °C reads 0.250 V. It is linear and the simplest temperature sensor to use. Check its supply range before you choose it: it needs roughly 4 V or more, so it is a poor fit for a board that runs at 3.3 V. Measuring below freezing needs extra circuitry.

The **DS18B20** sends temperature as digital data over a single wire, using a protocol called 1-Wire. It is accurate to about half a degree across the middle of its range, and you can choose between 9 and 12 bits of resolution. It will not communicate at all unless a 4.7 kΩ pull-up resistor connects the data line to the supply. Counterfeit chips are common and often less accurate, so buy from a reputable source.



Three ways to read temperature: a resistance, an analog voltage, and digital data.

A **thermocouple** is two different metals joined at a tip. It produces a tiny voltage, tens of microvolts per degree, that depends on the temperature difference between the tip and the board end. Reading that directly needs careful amplification and compensation for the board end's own temperature, which is why thermocouples are usually paired with an interface chip that reports the temperature digitally. Use one when other sensors would melt, such as furnaces, kilns and exhaust systems, where some probes work well above 1,000 °C. The right interface chip depends on your temperature range and supply voltage, so check its datasheet.

Light

A **photoresistor**, or LDR, drops in resistance as light increases. It is cheap and good for answering “is it light or dark?” It responds slowly, in tens of milliseconds, and it is not calibrated, so it cannot give you a trustworthy lux figure. Two LDRs with different part numbers can behave very differently.

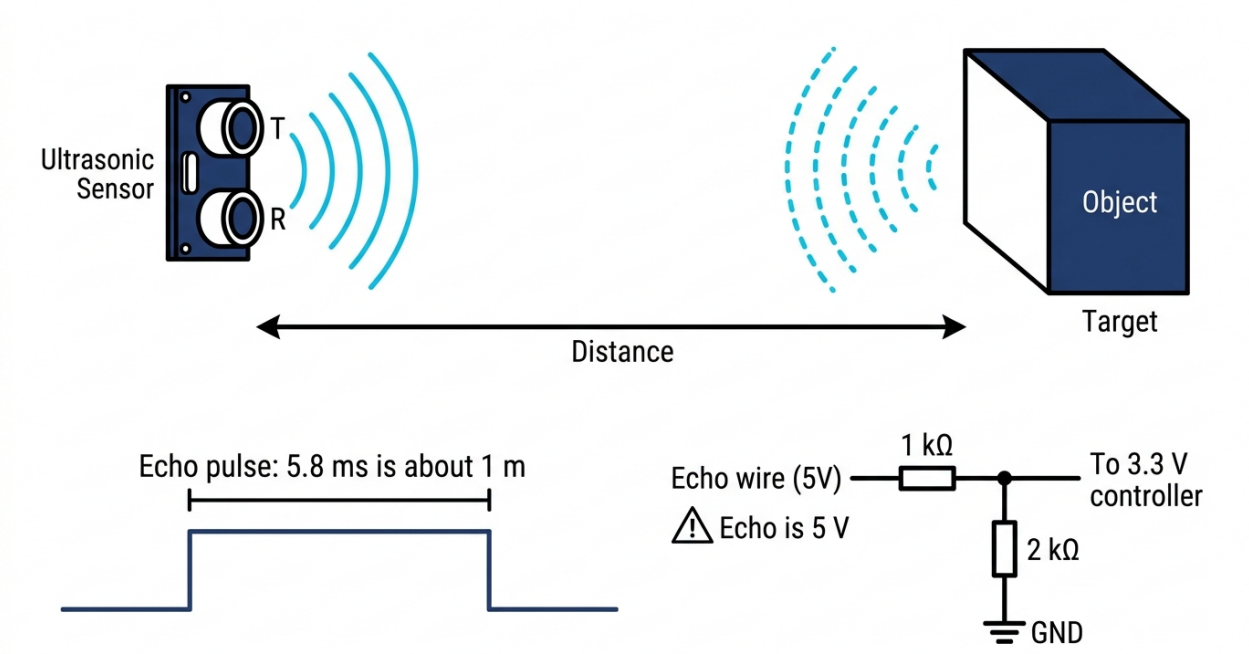
The **BH1750** reports illuminance in lux directly, as digital data over I^2C. Its resolution is about 1 lux, though its accuracy is only within tens of percent, which is plenty to tell bright from dim and not enough for lab work. The bare chip is a 3.3 V part, and many breakout boards add a regulator, so check what yours includes. A pin on the chip selects one of two addresses, so two sensors can share the same bus.

Distance

The **ultrasonic sensor** (the common HC-SR04 type) sends out a burst of sound and times the echo. Its output is a digital pulse whose width is proportional to the distance. Sound travels about 343 m/s in room-temperature air, and the pulse covers the trip out and back, so distance equals pulse time multiplied by 343, divided by 2. An echo pulse of 5.8 ms means about 1 m. The usable

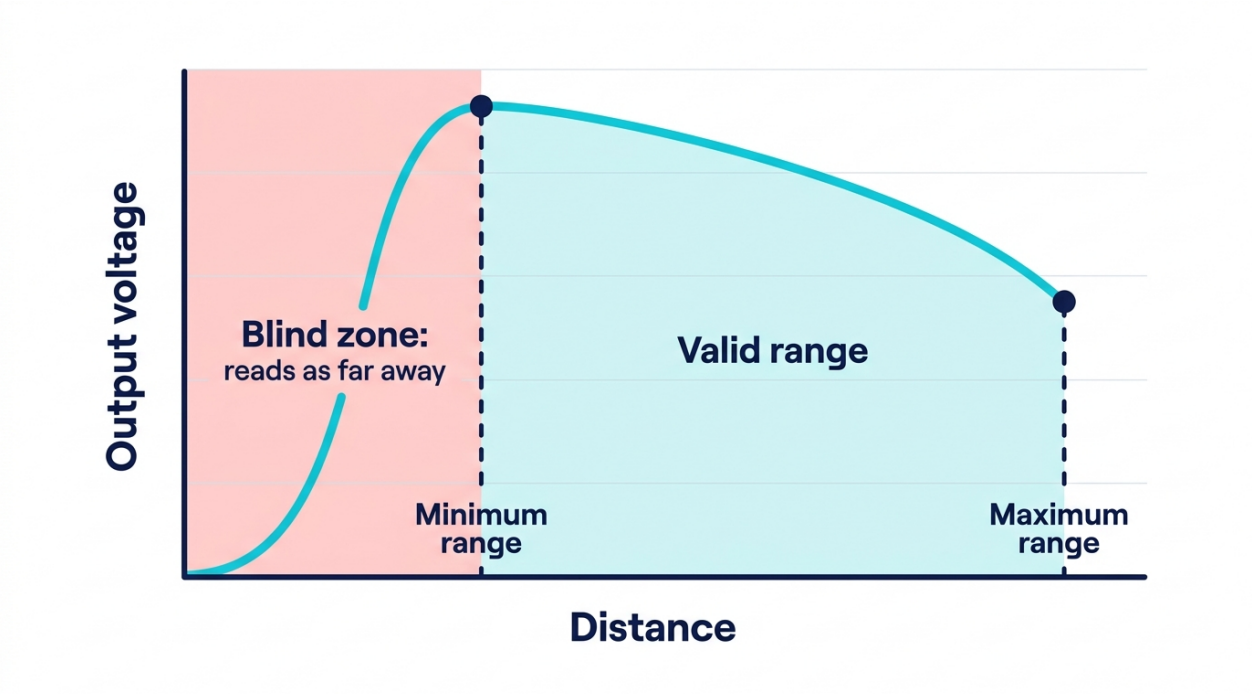
range is roughly 2 cm to 4 m. Closer than the minimum, readings are unreliable. Soft surfaces such as fabric and foam absorb the sound, and surfaces at a steep angle bounce it away from the sensor.

There is one wiring trap. This sensor runs on 5 V and its echo pin outputs 5 V, which can damage the pin of a 3.3 V controller. A divider of 1 kΩ and 2 kΩ brings it down to 3.33 V.



An ultrasonic sensor times an echo. Its 5 V echo pin needs a divider before a 3.3 V controller.

The **Sharp infrared distance sensor** (the GP2Y0A21YK0F type) outputs an analog voltage that falls as the object gets farther, over a range of roughly 10 to 80 cm. The curve is not a straight line, so you convert voltage to distance using a curve or formula fitted to the datasheet graph. Its trap is the minimum range. Inside it, the output drops, so an object placed very close can look as though it is far away.



Inside the minimum range the output voltage drops, so a very close object can look far away.

The **VL53L0X** is a laser time-of-flight sensor that reports distance in millimetres as digital data over I^2C. It reaches about 2 m. Its accuracy depends on how reflective the target is and on ambient light, so test it against your real target.

Motion and orientation

A **PIR sensor** (the HC-SR501 type) detects changes in infrared radiation, such as a warm body moving across its view. Its output is a plain on/off signal, and it reaches several metres across a wide cone. It takes about a minute to settle after power-up, and once triggered it holds its output for a set time. It detects movement, not a person standing still.

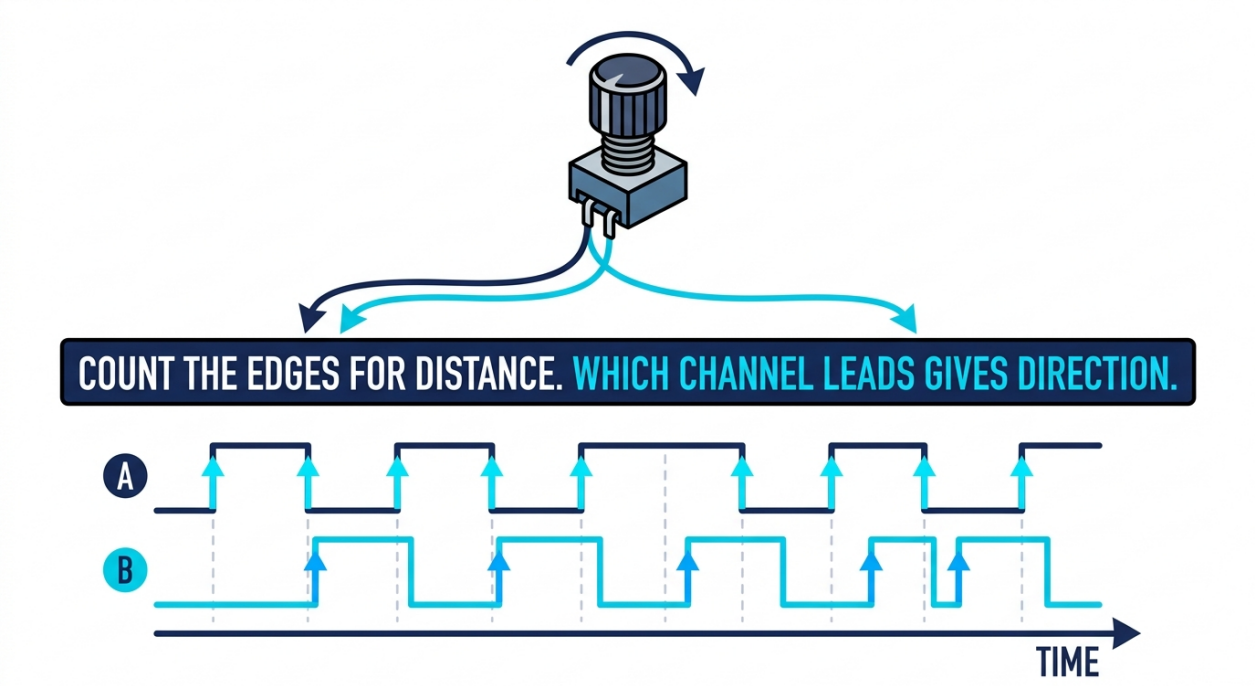
The **MPU-6050** is an inertial sensor that combines a three-axis accelerometer with a three-axis gyroscope, reporting digital data over I^2C. The ranges are selectable, from ±2 g to ±16 g for acceleration and from ±250 to ±2,000 degrees per second for rotation. The gyroscope reports how fast something is rotating, so you get an angle by adding up those readings over time, and tiny errors add up with them. That slow build-up is called drift. The accelerometer can give tilt from gravity but is noisy and gets confused when the device itself moves. Real systems combine the two with a filter. These sensors are widely used to watch tilt and vibration on machinery.

Position

A **potentiometer** gives an analog voltage that follows the shaft angle, which makes it a simple absolute angle sensor. It rotates roughly 300 degrees, with dead zones at the ends, and forcing it past its end stop can break the wiper.

An **incremental rotary encoder** outputs two square waves, A and B, offset slightly in timing. Counting the edges tells you how far the shaft has turned, and which channel leads tells you the direction. It reports change, not absolute position, so it forgets where it is when power is lost.

Mechanical encoders chatter as their contacts close, an effect called bounce, which causes missed or extra counts unless you filter it in hardware or software.



Encoder channels A and B. Count the edges for distance; which channel leads gives direction.

A **limit switch** gives a plain on/off signal when something reaches the end of its travel. Configure a pull-up or pull-down resistor, or the input will float and read random values. It also bounces for a few milliseconds when it closes, so debounce it before counting presses.

Hall effect sensors respond to magnetic fields. The A3144 is a digital type that switches when a magnet comes close, and it usually responds to one magnetic pole. Its output can only pull low, so it needs a pull-up resistor to produce a clean high. The SS49E is a linear type with an analog output that sits around the middle of its supply with no magnet and moves up or down with field strength and polarity. A magnet on a rotating shaft with a Hall sensor beside it is a common way to measure speed.

Force

A **force-sensitive resistor**, or FSR, drops in resistance as pressure rises. It is good for answering “was it pressed, and roughly how hard?” It is a poor weighing scale, because its response is not linear and varies with how the force is spread across it, with temperature, and over time.

A **load cell** is a metal bar carrying strain gauges that change resistance very slightly when it bends. That change is far too small to read directly, so a load cell is paired with an amplifier chip, the HX711 being the common one, which has a 24-bit ADC and sends digital data over its own two-wire interface. Despite having a clock pin and a data pin, it is not I^2C or SPI, so you need code written for it. Mounting matters, too. The load cell must be able to flex, so fix one end and leave a gap beneath the free end, or you will read zero. Load cells come rated for different capacities, from 1 kg to hundreds of kilograms, and are the standard choice for weighing.

Quick reference

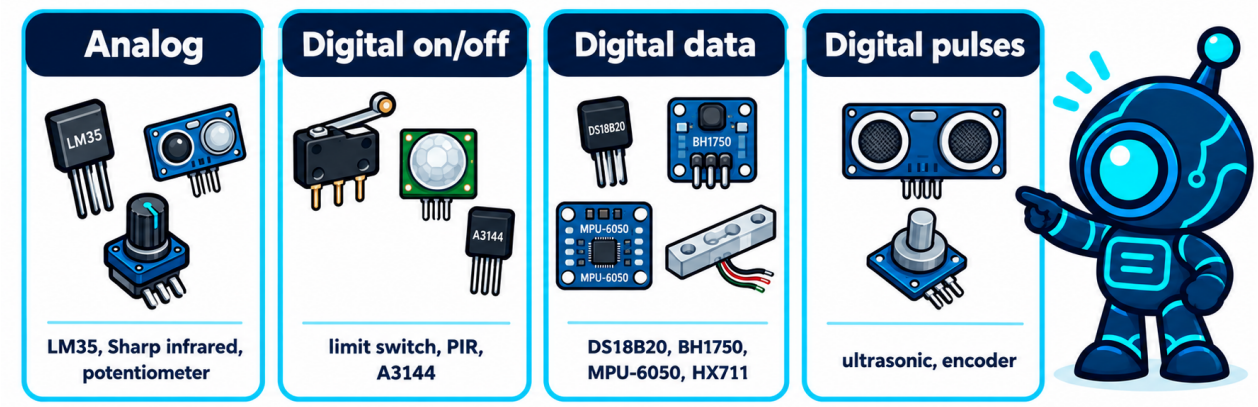
Sensor	Measures	Output	Watch out for
NTC thermistor	Temperature	Resistance (use a divider)	Non-linear, constants vary by part
LM35	Temperature	Analog, 10 mV per °C	Needs roughly 4 V or more
DS18B20	Temperature	Digital data, 1-Wire	4.7 kΩ pull-up, counterfeits
Thermocouple + interface chip	High temperature	Digital data, SPI	Choose chip for range and supply
Photoresistor (LDR)	Light level	Resistance (use a divider)	Slow, uncalibrated
BH1750	Light in lux	Digital data, I^2C	3.3 V part, address pin
Ultrasonic (HC-SR04)	Distance	Digital pulse width	5 V echo, soft and angled targets
Sharp infrared	Distance	Analog, non-linear	Wrong inside minimum range
VL53L0X	Distance	Digital data, I^2C	Target reflectivity, ambient light
PIR (HC-SR501)	Motion	Digital on/off	Warm-up, hold time
MPU-6050	Acceleration, rotation	Digital data, I^2C	Gyroscope drift
Potentiometer	Angle	Analog	Dead zones, end stops
Rotary encoder	Rotation, direction	Digital pulses (A and B)	Bounce, relative only
Limit switch	Contact	Digital on/off	Pull-up, bounce
Hall effect (A3144, SS49E)	Magnetic field	On/off or analog	A3144 needs a pull-up
FSR	Force, pressure	Resistance (use a divider)	Not for accurate weight
Load cell + HX711	Weight	Digital data, two-wire	Mounting must allow flex

A few sensors fit none of the three signal types from Topic 3.1 cleanly. The ultrasonic sensor and the encoder output digital pulses whose timing or count carries the measurement. You read them by timing or counting pulses, not by reading a level.

Choosing a sensor

Start with the quantity. Then check the range you need. Then match the output to what your controller can read: does it have an ADC, what logic voltage does it use, and how many free pins? Finally, read the datasheet for the exact part, as you learned in Topic 2.4.

Consider a conveyor that must stop when a box reaches the end. You only need a yes or no, so a digital on/off sensor is enough, such as a limit switch or a distance sensor with a set threshold. If instead you need to know where the box is along the belt, an encoder on the belt motor counts how far the belt has moved, which tells you the position.



The same sensors sorted by output type: analog, digital on/off, digital data and digital pulses.

> ***Going deeper:*** *the Robotics course revisits these sensors for machines that must perceive their surroundings and combine several readings at once, including distance and motion sensors in more depth.*

Where this leaves you

You can now sort almost any sensor into a family by what it measures and into a signal type by what it outputs. You know the wiring traps that damage hardware or silently ruin readings: a missing pull-up, a 5 V echo on a 3.3 V pin, a gyroscope that drifts, a load cell mounted flat. Next comes Topic 3.3, where we connect these to a microcontroller and write the code that reads an analog voltage, a digital pin and an I^2C sensor.